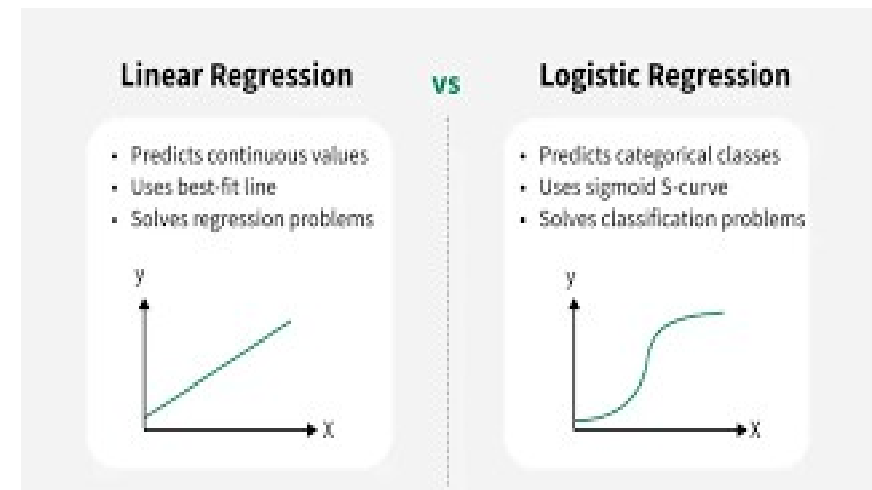


Classification metrics Clinic

- Varun Raste

Logistic Regression

- **Probabilistic Classifier:** Outputs probability of the positive class and classifies using a threshold (default = 0.5).
- **Interpretable Coefficients:** Log-odds representation helps understand feature impact.
- **Linear Decision Boundary:** Performs best when classes are linearly separable.
- **Sensitive to Threshold:** AUC-ROC and precision–recall curves help evaluate across thresholds.
- **Regularization:** L1/L2 penalties prevent overfitting and stabilise coefficients.



Precision & Recall

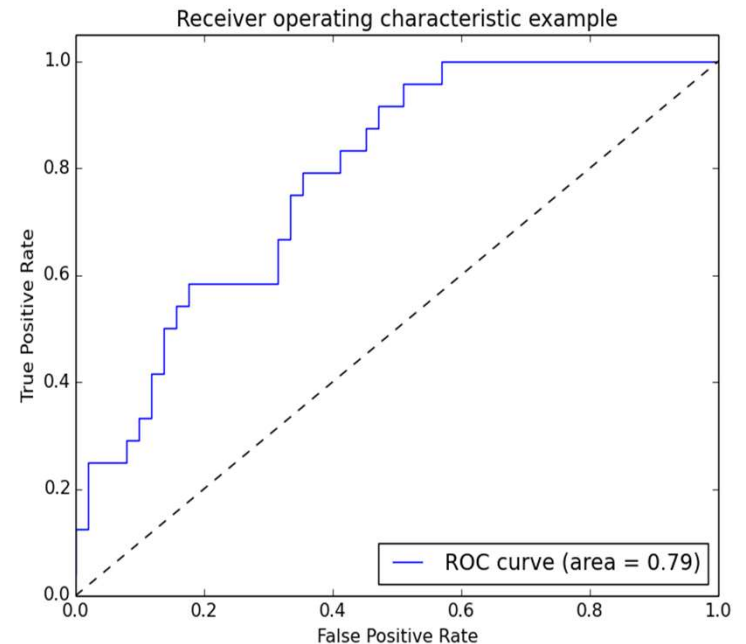
- **Precision = $TP / (TP + FP)$:**
Measures accuracy of positive predictions.
 - **Useful When False Positives Are Costly:**
e.g., fraud alerts, disease diagnosis.
 - **High Precision = Few Wrong Positives:**
Model is conservative when predicting positives.
 - **Improves with Higher Threshold:** Predict “positive” only when highly confident.
 - **Trade-off:** Often decreases recall.
- **Recall = $TP / (TP + FN)$:**
Measures ability to capture actual positives.
 - **Useful When Missing Positives Is Expensive:** e.g., cancer detection, safety systems.
 - **High Recall = Few Missed Positives:** Model catches most true cases.
 - **Increases with Lower Threshold:** More positives predicted → more detected.
 - **Trade-off:** Precision may drop.

F1-Score & Accuracy

- **Harmonic Mean of Precision & Recall:**
$$F1 = 2 * (P * R) / (P + R).$$
 - **Useful in Imbalanced Datasets:** Balances false positives and false negatives.
 - **Insensitive to True Negatives:** Focuses purely on positive-class quality.
 - **Ideal When P & R Are Both Important:**
Good for binary classification trade-offs.
 - **Range:** 0 to 1 (higher = better).
- **Accuracy = (TP + TN) / Total:**
Measures overall correctness of predictions.
 - **Misleading on Imbalanced Data:** Looks high even if minority class is ignored.
 - **Best When Classes Are Balanced:** Works well for symmetric classification tasks.
 - **Simple to Interpret:** Stakeholder-friendly, but not robust.
 - **Should Not Be Sole Metric:** Combine with precision/recall for clarity

ROC AUC Curve

- **ROC Curve:** Plots True Positive Rate vs False Positive Rate across different classification thresholds.
- **Threshold-Independent View:** Shows how a model performs across all possible cutoffs, not just at 0.5.
- **AUC (Area Under Curve):** Single metric summarizing ROC — higher AUC means better ability to separate classes.
- **Interpretability:** AUC = probability that the model ranks a random positive case higher than a random negative one.
- **Usage:** Ideal when classes are balanced or when you care about ranking quality more than calibrated probabilities.



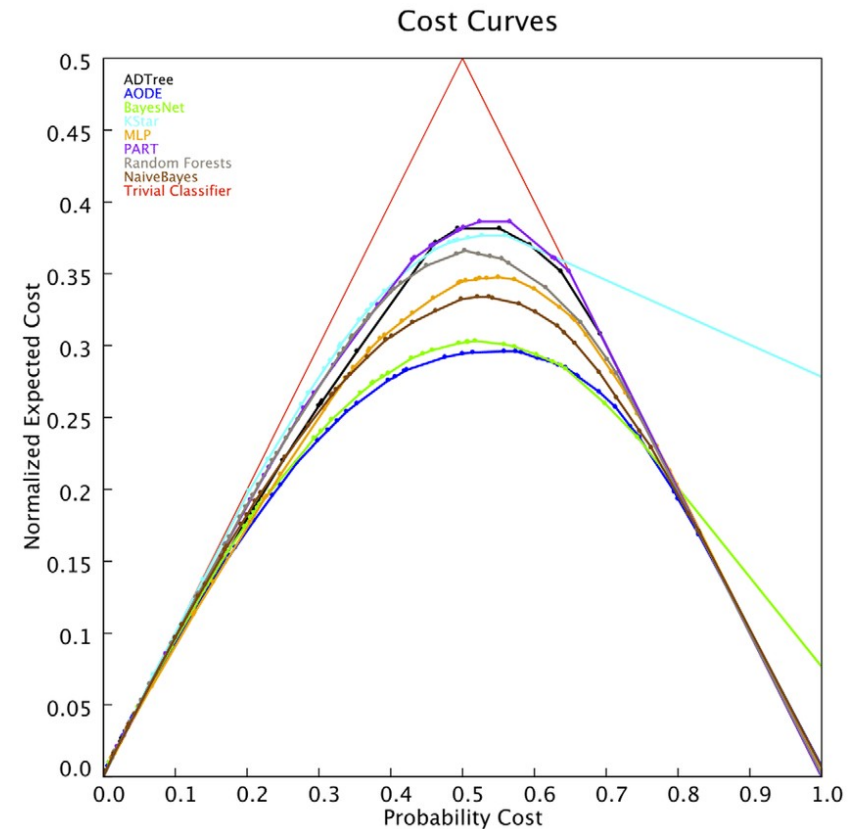
Confusion Matrix

- **4-Quadrant Summary:** Shows TP, TN, FP, FN — giving complete insight into prediction outcomes.
- **Direct Error Visibility:** Makes it easy to see which error is dominating (false positives vs false negatives).
- **Metric Foundation:** Forms the basis for precision, recall, F1, accuracy, specificity, etc.
- **Custom Error Costs:** Helps apply domain-specific penalties on FP and FN for cost-sensitive evaluation.
- **Diagnostic Tool:** Highlights class imbalance issues and model bias at the chosen threshold.

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

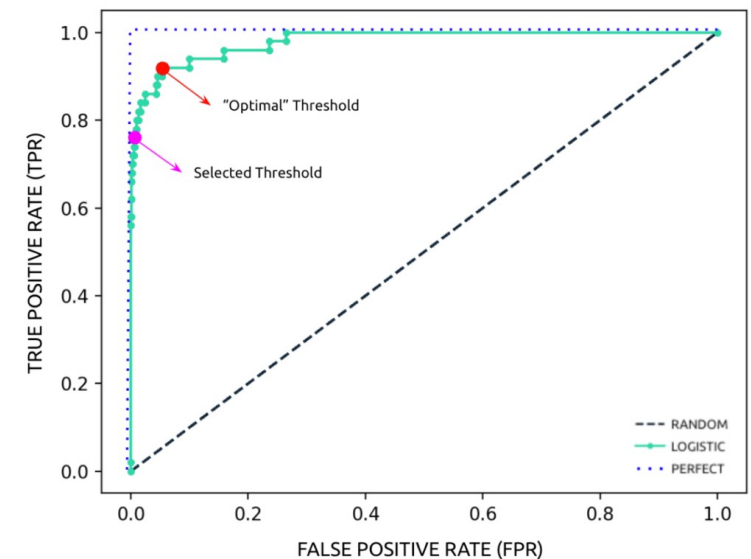
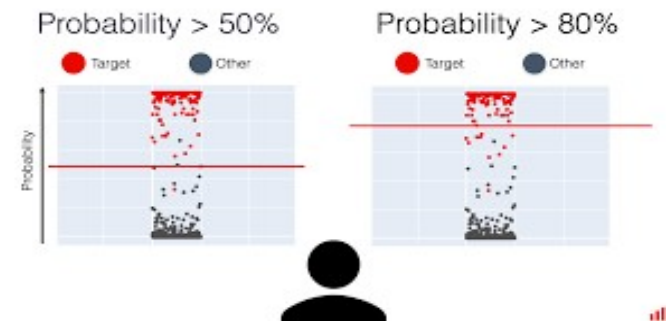
Cost Curves

- **Visualizes Expected Cost:** Shows how the total cost of predictions changes as the classification threshold shifts.
- **Incorporates FP & FN Costs:** Unlike ROC, it directly embeds business costs (e.g., fraud misses vs false alerts).
- **Threshold Optimization:** The lowest point on the curve indicates the threshold that minimizes overall expected loss.
- **Model Comparison:** Multiple curves can be plotted to see which model performs best under different cost scenarios.
- **Better for High-Stakes Decisions:** Ideal when error costs are asymmetric or the dataset is highly imbalanced.



Threshold Tuning

- **Classification Threshold Controls P/R Trade-off:** Adjusting threshold shifts precision vs recall.
- **ROC & PR Curves Help Selection:** Choose threshold based on business cost.
- **Custom Threshold > Default 0.5:** Often necessary in skewed datasets.
- **Optimize for Metric:** E.g., maximize F1 or minimize expected cost.
- **Business-driven Triggering:** Align threshold with operational risk appetite.



Consequences of Imbalanced data

- **Misleading Accuracy:** A model can show very high accuracy simply by predicting the majority class, while completely ignoring the minority class.
- **Poor Minority Class Detection:** The model tends to fail at identifying rare but critical cases (fraud, disease, churn), leading to high false negatives.
- **Biased Decision Boundary:** The classifier shifts its boundary toward the majority class, causing underfitting of the minority patterns.
- **Unreliable Metrics:** Standard metrics like accuracy and ROC-AUC overestimate performance; precision–recall metrics become more relevant.
- **Model Instability:** Small changes in data can drastically alter predictions due to weak representation of minority examples.

Handling Imbalanced data

- **Resampling:** Oversampling minority (SMOTE) or undersampling majority.
- **Class Weights:** Penalize errors on minority class more heavily.
- **Synthetic Data Methods:** Advanced SMOTE variants for better minority representation.
- **Threshold Shifting:** Lower threshold to improve recall on minority class.
- **Use PR Curve > ROC:** Better reflects performance in skewed datasets

